

Pregnancy outcomes in inflammatory bowel disease patients treated with vedolizumab, anti-TNF or conventional therapy: results of the European CONCEIVE study.

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Women with inflammatory bowel diseases (IBD) often receive biologicals during pregnancy to maintain disease remission. Data on outcome of vedolizumab-exposed pregnancies (VDZE) are sparse. To assess pregnancy and child outcomes of VDZE pregnancies and to compare these results to anti-TNF exposed (TNFE) or both immunomodulatory and biologic unexposed (CON IBD) pregnancies. A retrospective multicentre case-control observational study was performed. VDZE group included 79 pregnancies in 73 IBD women. The TNFE and CON IBD group included 186 pregnancies (162 live births) in 164 IBD women and 184 pregnancies (163 live births) in 155 IBD women, respectively. At conception, cases more often had active disease ([VDZE: 36% vs TNFE: 17%, $P = .002$] and [VDZE: 36% vs CON IBD: 24%, $P = .063$]). No significant difference in miscarriage rates were found between groups (VDZE and TNFE: 16% vs 13%, $P = .567$; VDZE and CON IBD: 16% vs 10%, $P = .216$). In live-born infants, median gestational age and birthweight were similar between groups. Median Apgar score at birth was numerically equal. Prematurity was similar in the VDZE group compared to the control groups, even when correcting for disease activity during pregnancy. The frequency of congenital anomalies was comparable between groups as were the percentages of breastfed babies. During the first year of life, no malignancies were reported and infants' infection risk did not significantly differ between groups. No new safety signal was detected in VDZE pregnancies although larger, prospective studies are required for confirmation.

Treating Inflammatory Bowel Disease in Pregnancy: The Issues We Face Today.

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Many women of childbearing age are living with inflammatory bowel disease [IBD], yet there are limited studies on the use of IBD medications in pregnancy. In this review, we provide a comprehensive update on the safety of these medications during pregnancy, particularly thiopurines and biologicals. Antibiotics, steroids, and aminosalicylates are relatively low risk for use in pregnancy, and growing evidence supports the safety of immunomodulators and anti-tumour necrosis factor agents as well. Available studies on infliximab, adalimumab, and certolizumab pegol show no increase in adverse events during pregnancy or perinatally. Similarly, studies on lactation demonstrate that concentrations of subcutaneous anti-tumour necrosis factor biologicals are undetectable, and levels of thiopurines and infliximab are negligible in breast milk. Less is known about anti-integrins in pregnancy [eg natalizumab and vedolizumab] but currently available data suggest they may be safe as well. Although more studies are needed to examine the long-term effects of these medications on offspring, the available data provide reassuring information for providers caring for women of childbearing age.

Safety of immunomodulators and biologics for the treatment of inflammatory bowel disease during pregnancy and breast-feeding.

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The aim of this article is to critically review available data regarding the safety of immunomodulators and biological therapies during pregnancy and breast-feeding in women with inflammatory bowel disease.

Methotrexate and thalidomide can cause congenital anomalies and are contraindicated during pregnancy (and breast-feeding). Although thiopurines have a Food and Drug Administration (FDA) rating D, available data suggest that these drugs are safe and well tolerated during pregnancy. Although traditionally women receiving azathioprine or mercaptopurine have been discouraged from breast-feeding because of theoretical potential risks, it seems that these drugs may be safe in this scenario. Treatment with cyclosporine for steroid-refractory ulcerative colitis (UC) during pregnancy can be considered safe and effective, and the use of this drug should be considered in cases of severe UC as a means of avoiding urgent surgery. Breast-feeding is contraindicated for patients receiving cyclosporine. Biological therapies appear to be safe in pregnancy, as no increased risk of malformations has been demonstrated. Therefore, the limited clinical results available suggest that the benefits of infliximab and adalimumab in attaining response and maintaining remission in pregnant patients might outweigh the theoretical risks of drug exposure to the fetus. Stopping therapy in the third trimester may be considered, as it seems that transplacental transfer of infliximab is low prior to this. Certolizumab differs from infliximab and adalimumab in that it is a Fab fragment of an antitumor necrosis factor alpha monoclonal antibody, and therefore it may not be necessary to stop certolizumab in the third trimester. The use of infliximab is probably compatible with breast-feeding.

Biologics During Pregnancy and Breastfeeding Among Women With Rheumatic Diseases: Safety Clinical Evidence on the Road.

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Females are generally more affected by autoimmune diseases, a fact that underlines the relationship with pregnancy and the safety of anti-rheumatic drugs in pregnancy and lactation. Biologic therapies are increasingly prescribed to treat and maintain remission in a significant number of systemic autoimmune rheumatic diseases. The experience with the use of biologics during gestation is extremely lacking because of the observational nature of the available studies and the difficulty in designing proper clinical trials in pregnancy. Among the studied biologics, more information was published on TNF α inhibitors and, in particular, on their potential passage through the placenta and impact on the fetus. Currently, a fragment of anti-TNF α monoclonal IgG, certolizumab pegol, is considered safe with almost no placental transfer. Subsequent observations are suggesting a comparable safety for the soluble TNF α receptor etanercept. Another biologic, eculizumab, the anti-C5a antibody used to treat complement-mediated microangiopathies, is also considered safe due to the unique engineered IgG2/4k formulation that limits its passage through the placental barrier. Still, long-term data about children born to women treated with biologics in pregnancy are not attainable. Data on breastfeeding are currently available for several biologics. This article reviews the literature available about which drugs are considered safe during pregnancy and lactation, which are not, and on future prospects.

Transplacental transfer of immunosuppressants and biologics used for the treatment of inflammatory bowel disease.

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The aim of the present article was to review the role of the placenta function in the transfer of immunosuppressants and monoclonal antibodies used for the treatment of inflammatory bowel disease,

as well as the available data regarding the safety of these therapies during pregnancy. Methotrexate can cause congenital abnormalities and is contraindicated during pregnancy. Available data suggest that thiopurines are safe and well tolerated during pregnancy. Treatment with cyclosporine for steroid refractory ulcerative colitis during pregnancy can be considered effective and safe. Biological therapies appear to be safe during pregnancy, as no increased risk of malformations has been demonstrated. Transplacental exchanges are known to involve passive transfer, active transport and facilitated diffusion. Thiopurines are transported by passive transfer. The transport across the placenta of corticosteroids, methotrexate and cyclosporine is modulated by protein "pumps". Infliximab and adalimumab bind the FcRn-receptor as other IgG proteins do. The mechanisms of the diffusion of certolizumab across the placenta barrier remain unknown.

[Biological therapy in women with inflammatory bowel disease during pregnancy].

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The aim of this article is to objective review available research data regarding the safety of biological therapies during pregnancy and breastfeeding in women with inflammatory bowel disease. Biological therapies appear to be safe in pregnancy, as no increased risk of malformations has been demonstrated. Available clinical results suggest that the efficacy of infliximab and adalimumab in achieving clinical response and maintaining remission in pregnant patients might outweigh the theoretical risks of drug exposure to the fetus. If possible, anti-TNF therapy should be stopped by the end of the second trimester due to transplacental transfer and potential risk for the fetus. The use of infliximab and adalimumab is probably compatible with breastfeeding.

Appropriateness of small molecule agents for patients with IBD of childbearing age - a RAND/UCLA appropriateness panel.

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Many women of childbearing age with inflammatory bowel disease (IBD) require advanced therapies. While biologics are largely low risk during pregnancy, the novel small molecules tofacitinib, filgotinib, upadacitinib and ozanimod (TFUO) have shown concerning teratogenic effects, and decreased fertility in animal studies. Therefore, their use in women of childbearing age needs careful consideration. RAND/University of California Los Angeles (UCLA) Appropriateness Method (RAM). To evaluate the appropriateness of TFUO in women of childbearing age. We convened a panel of six gastroenterologists, two IBD nurses, one IBD pharmacist and three expert patients. Following a literature review, 13 statements were drafted and voted upon in 2 rounds. All 13 statements were deemed appropriate. The panel concluded that women with IBD of childbearing age who wish to commence therapy with TFUO, need to use effective contraception and be counselled regarding the risk in unplanned pregnancies. For women using contraception while on Janus kinase inhibitor (JAKi) therapy, we suggest the preferred use of progesterone-only or non-hormonal long-acting contraception. TFUO are contraindicated during pregnancy and breast feeding. We recommend that women receiving TFUO cease therapy in time to establish clinical remission for at least 3 months prior to conception. Therapies other than TFUO should be considered as first-line therapy in women with IBD of childbearing age, except in select individual circumstances. TFUO may be appropriate for women of childbearing age after failure of, intolerance or

contraindications to one biological agent. TFUO should be avoided during pregnancy and breastfeeding, and alternative therapies should be considered as first-line treatments. We provide clinical practice recommendations regarding the use of TFUO for IBD in women of childbearing age.

Management of asthma in pregnancy.

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Asthma is the most common chronic disease to affect pregnant women and can have a significant effect on pregnancy outcomes, with increased rates of preterm birth, premature delivery and caesarean section observed if poorly controlled. Pregnancy can also influence asthma control. Prescribing in pregnancy causes anxiety for patients and healthcare professionals and can result in alteration or undertreatment of asthma. Good asthma control with prompt and adequate management of exacerbations is key to reducing adverse pregnancy outcomes for both mother and fetus. The majority of asthma treatment can be continued as normal in pregnancy and there is emerging evidence of the safety of biologic medications also. This article aims to summarise the current evidence about asthma in pregnancy and guide the appropriate management of this population.

Biologics Use for Psoriasis during Pregnancy and Its Related Adverse Outcomes in Pregnant Women and Newborns: Findings from WHO Pharmacovigilance Study.

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The safety of biologics, other than TNF- α inhibitors, during pregnancy has not been sufficiently established. To assess the risk of pregnancy-related adverse outcomes of biologics used for psoriasis, compared to TNF- α inhibitors, we utilized the WHO global pharmacovigilance database (1968-2024). We utilized the World Health Organization's global pharmacovigilance database from 1968 to 2024. From over 140 million reports from more than 170 countries, we extracted 6,518 reports of pregnancy-related adverse outcomes associated with the biologics of interest. These biologics included TNF- α inhibitors (e.g., etanercept, infliximab, adalimumab, certolizumab pegol), IL-12/IL-23 inhibitor (e.g., ustekinumab), IL-17 inhibitors (e.g., secukinumab, brodalumab, ixekizumab, bimekizumab), and IL-23 inhibitors (e.g., guselkumab, tildrakizumab, risankizumab). Each biologic was compared to TNF- α inhibitors and certolizumab pegol in two separate disproportionality analyses. The reporting odds ratio (ROR) was calculated for maternal, fetal, and neonatal outcomes, categorized into seven major groups. Multivariable and sensitivity analyses were conducted to validate the primary results. The disproportionality analysis showed that, compared to TNF- α inhibitors, most biologics had a lower frequency of pregnancy-related adverse outcomes, with the exception of brodalumab. Specifically, ROR and 95% confidence intervals (CIs) were as follows: ustekinumab (ROR, 0.27; 95% CI: 0.21-0.35), secukinumab (0.17; 0.13-0.22), brodalumab (0.20; 0.02-2.21), ixekizumab (0.05; 0.03-0.08), bimekizumab (0.10; 0.01-0.71), guselkumab (0.09; 0.05-0.15), tildrakizumab (0.02; 0.00-0.14), and risankizumab (0.38; 0.25-0.58). However, risankizumab was reported with a higher frequency of abortion and stillbirth (1.87; 1.32-2.63). These findings were consistent when compared to certolizumab pegol, as well as in multivariable and sensitivity analyses. Furthermore, when comparing other TNF- α inhibitors to certolizumab pegol, infliximab showed a lower frequency of pregnancy-related adverse outcomes (ROR, 0.71; 95% CI: 0.55-0.92), etanercept showed a comparable frequency (1.00; 0.77-1.31), and adalimumab showed a higher frequency (1.42;

1.11-1.81). Biologics used for psoriasis, with the exception of brodalumab, exhibit a lower frequency of pregnancy-related adverse outcomes compared to TNF- α inhibitors and certolizumab pegol, suggesting their potential to be safe options during pregnancy. However, further studies are necessary to evaluate the safety of these biologics during pregnancy, accounting for confounding factors.

Navigating Reproductive Care in Patients With Inflammatory Bowel Disease: A Comprehensive Review.

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Inflammatory bowel disease [IBD] is often diagnosed in patients during their reproductive years. It is crucial that both healthcare providers and patients are adequately informed to avoid misguided decisions regarding family planning. One of the most important aspects during conception and pregnancy is to maintain disease remission, as disease activity is associated with adverse pregnancy outcomes. Apart from methotrexate, most conventional drugs used in IBD are considered low risk during conception and pregnancy. For newer agents, evidence is still limited. If needed, surgery must not be postponed and should ideally be performed in specialized centres. In most patients, delivery should be vaginal except for patients with complex perianal disease, with an ileoanal pouch anastomosis, or if there is an obstetric contraindication. In children exposed to biological treatments during pregnancy, the risk of infections appears to be low, and psychomotor development is probably not affected. Regarding immunizations, the standard vaccination schedule for inactivated vaccines should be followed for children exposed to biologics in utero. In the case of live vaccines, such as rotavirus, decisions should be individualized and take into consideration the risk-benefit ratio, particularly in developing countries. In this review, we provide a comprehensive and updated overview of aspects related to fertility, pregnancy, breastfeeding, and the impact on the care of children born to mothers with IBD. Both the available evidence and areas of uncertainty are discussed, with the goal of assisting healthcare professionals caring for IBD patients during this important stage of their lives.